

## Instalación de Codex y Archify

Capturas y prompt en el orden del archivo original, con una explicación breve de cada paso.

## Codex

```

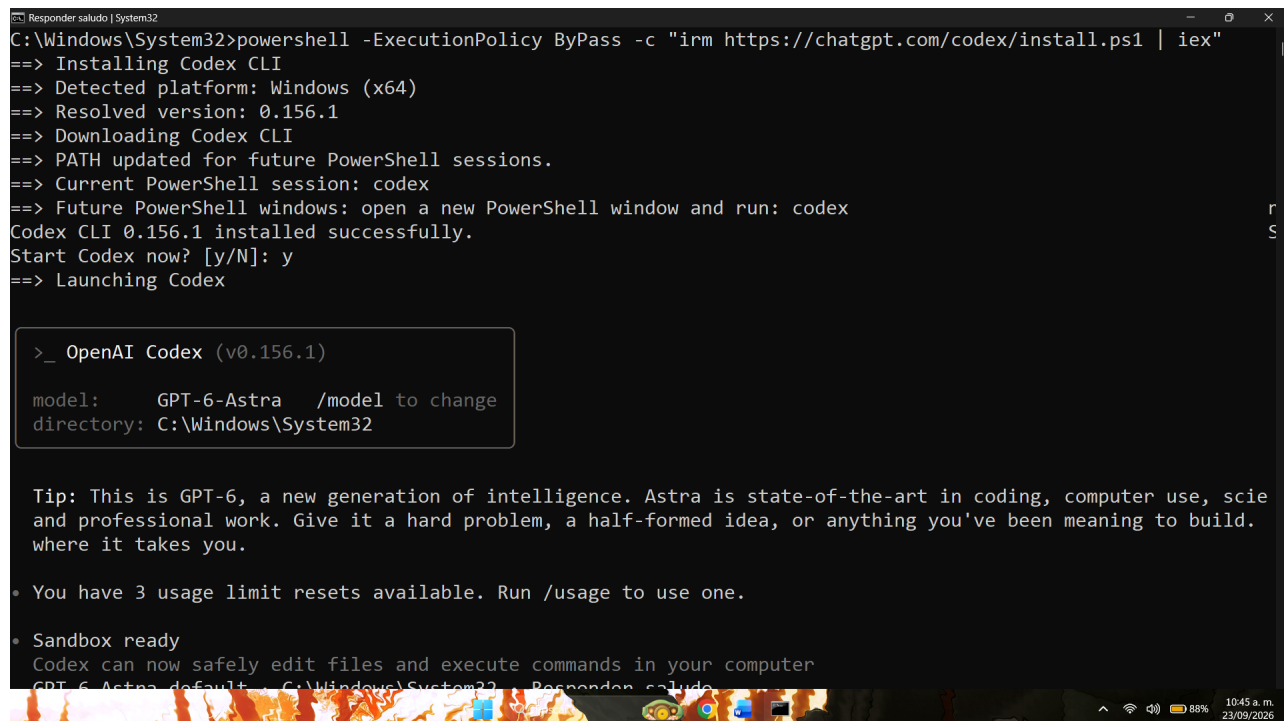
C:\Windows\System32>winget install Python.Python.3.13
Found Python 3.13 [Python.Python.3.13] Version 3.13.15
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://www.python.org/ftp/python/3.13.15/python-3.13.15-amd64.exe
[REDACTED] 28.0 MB / 28.0 MB
Successfully verified installer hash
Starting package install...
Successfully installed

C:\Windows\System32>py --version
Python 3.13.15

C:\Windows\System32>powershell -ExecutionPolicy Bypass -c "irm https://chatgpt.com/codex/install.ps1 | iex"
==> Installing Codex CLI
==> Detected platform: Windows (x64)
==> Resolved version: 0.156.1
==> Downloading Codex CLI

```

Captura 01. Se instala Python y se ejecuta desde CMD el comando de PowerShell que inicia la instalación de Codex CLI.



```
C:\Windows\System32>powershell -ExecutionPolicy ByPass -c "irm https://chatgpt.com/codex/install.ps1 | iex"
==> Installing Codex CLI
==> Detected platform: Windows (x64)
==> Resolved version: 0.156.1
==> Downloading Codex CLI
==> PATH updated for future PowerShell sessions.
==> Current PowerShell session: codex
==> Future PowerShell windows: open a new PowerShell window and run: codex
Codex CLI 0.156.1 installed successfully.
Start Codex now? [y/N]: y
==> Launching Codex

> _ OpenAI Codex (v0.156.1)

model:      GPT-6-Astra  /model to change
directory:  C:\Windows\System32

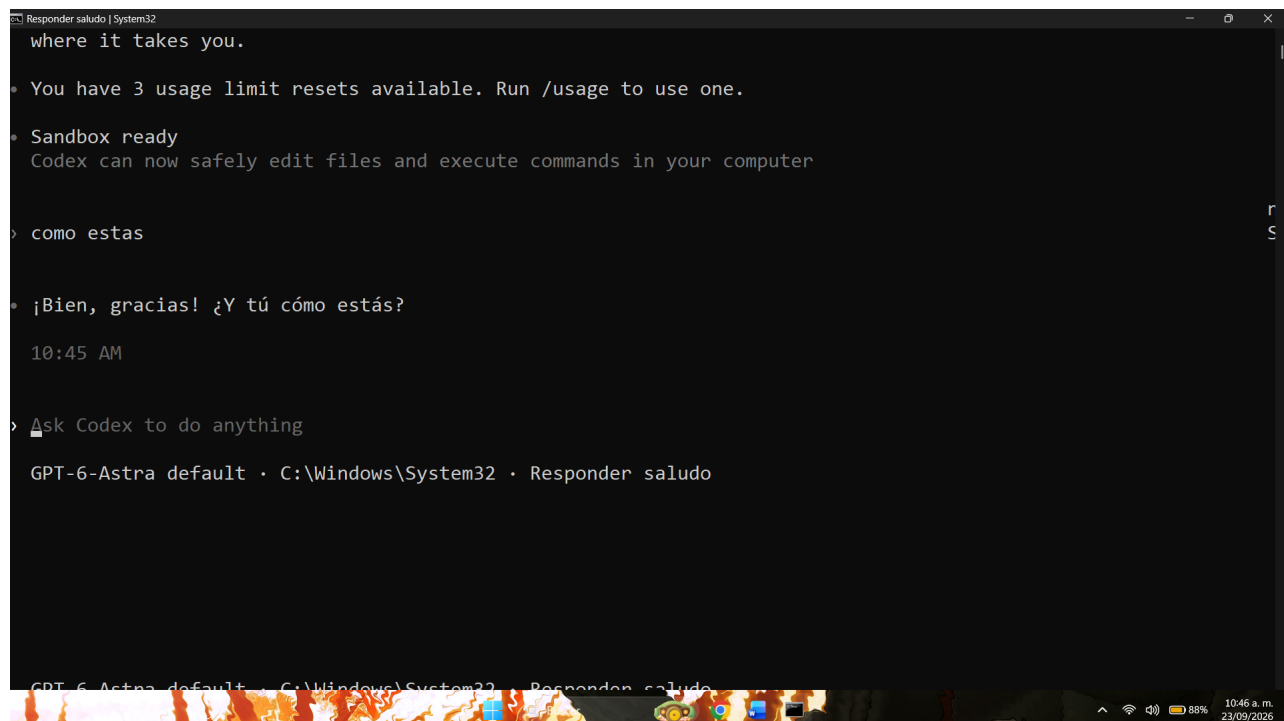
Tip: This is GPT-6, a new generation of intelligence. Astra is state-of-the-art in coding, computer use, science and professional work. Give it a hard problem, a half-formed idea, or anything you've been meaning to build, where it takes you.

• You have 3 usage limit resets available. Run /usage to use one.

• Sandbox ready
Codex can now safely edit files and execute commands in your computer
GPT-6-Astra default • C:\Windows\System32 • Responder saludo
```

Captura 02. La terminal confirma la instalación de Codex CLI y abre su interfaz interactiva.

## Prueba de instalación



```
where it takes you.

• You have 3 usage limit resets available. Run /usage to use one.

• Sandbox ready
Codex can now safely edit files and execute commands in your computer

> como estas

• ¡Bien, gracias! ¿Y tú cómo estás?

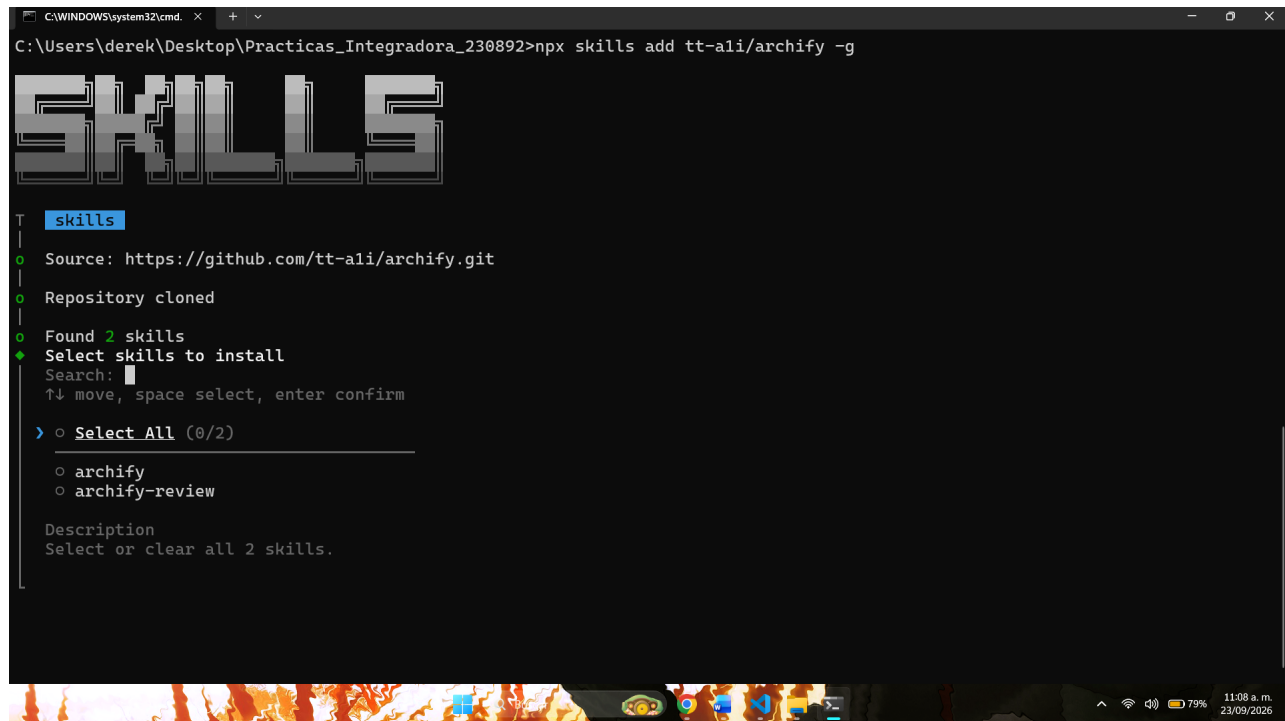
10:45 AM

> Ask Codex to do anything

GPT-6-Astra default • C:\Windows\System32 • Responder saludo
```

Captura 03. Se escribe un mensaje de prueba en español y Codex responde. Esto comprueba que la sesión quedó operativa.

# Archify

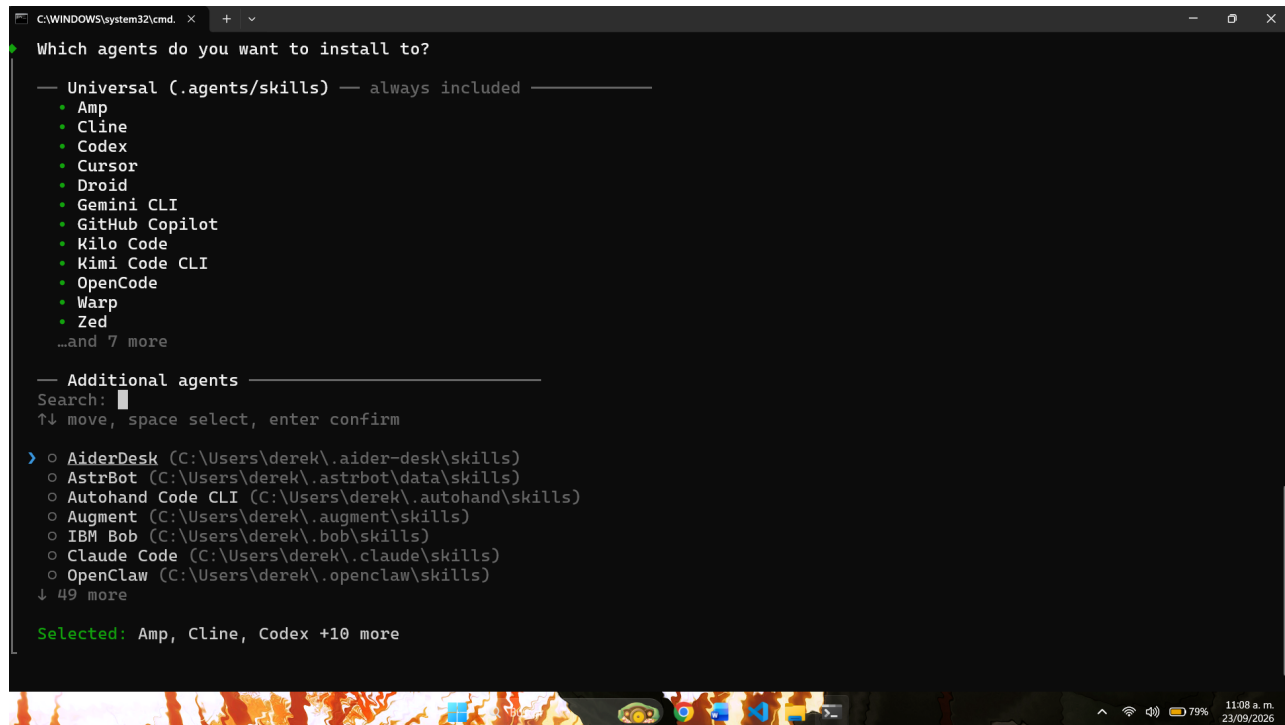


```
C:\WINDOWS\system32\cmd. x + v
C:\Users\derek\Desktop\Practicas_Integradora_230892>npx skills add tt-ali/archify -g

SKILLS

T skills
o Source: https://github.com/tt-ali/archify.git
o Repository cloned
o Found 2 skills
o Select skills to install
  Search:
  ↑↓ move, space select, enter confirm
  > o Select All (0/2)
    o archify
    o archify-review
  Description
  Select or clear all 2 skills.
```

Captura 04. Desde la carpeta del proyecto se ejecuta la instalación global de Archify. El asistente encuentra archify y archify-review; la barra espaciadora marca las opciones y Enter confirma.



```
C:\WINDOWS\system32\cmd. x + v
• Which agents do you want to install to?

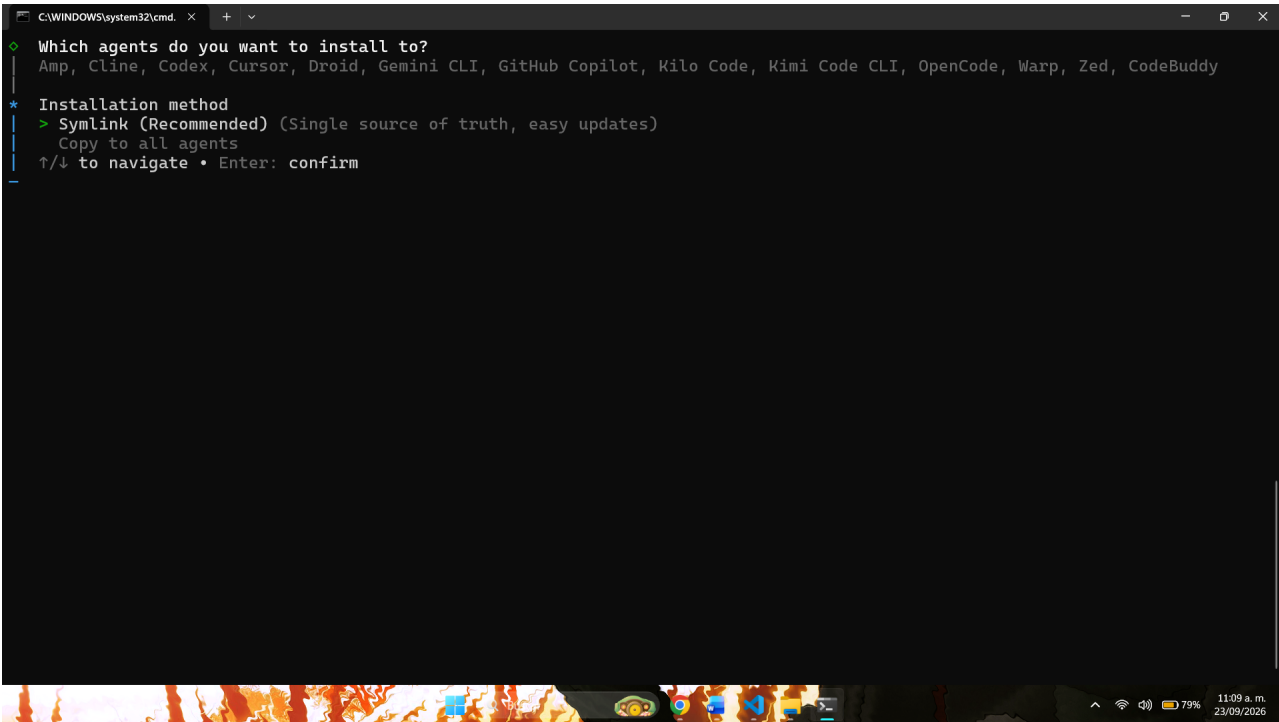
— Universal (.agents/skills) — always included —
• Amp
• Cline
• Codex
• Cursor
• Droid
• Gemini CLI
• GitHub Copilot
• Kilo Code
• Kimi Code CLI
• OpenCode
• Warp
• Zed
...and 7 more

— Additional agents —
Search:
↑↓ move, space select, enter confirm

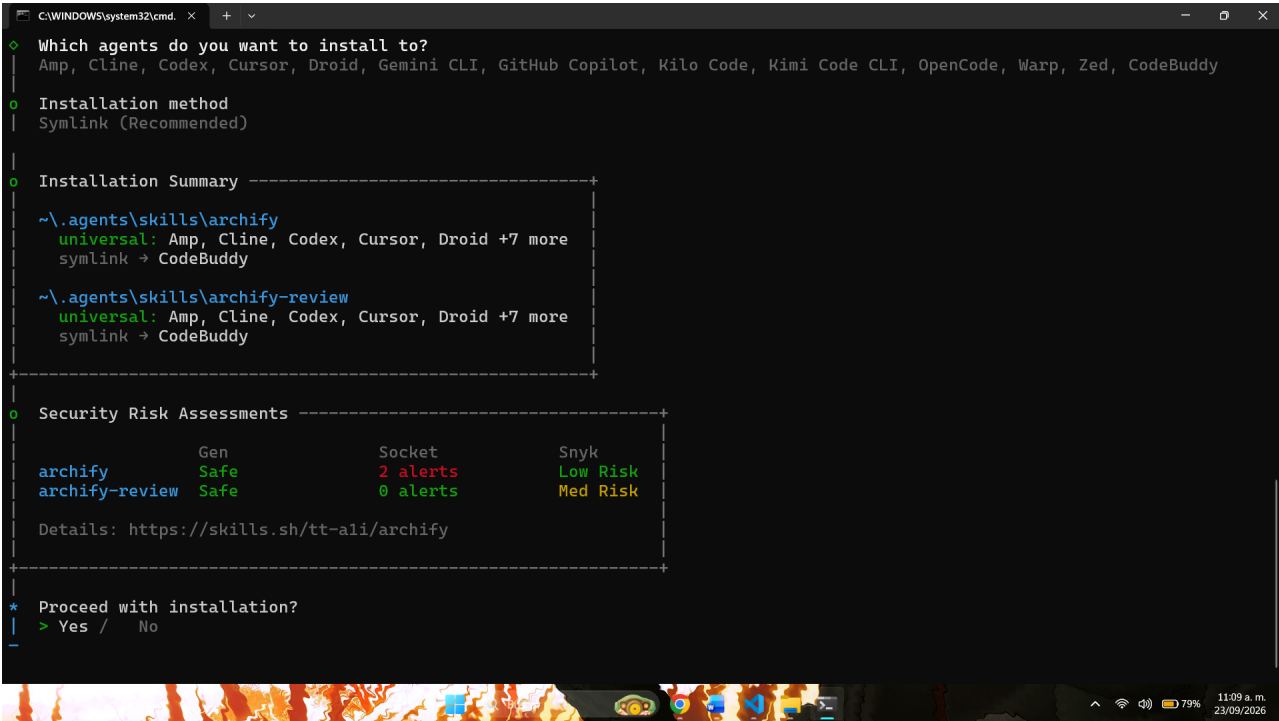
> o AiderDesk (C:\Users\derek\.aider-desk\skills)
  o AstrBot (C:\Users\derek\.astrbot\data\skills)
  o Autohand Code CLI (C:\Users\derek\.autohand\skills)
  o Augment (C:\Users\derek\.augment\skills)
  o IBM Bob (C:\Users\derek\.bob\skills)
  o Claude Code (C:\Users\derek\.claude\skills)
  o OpenClaw (C:\Users\derek\.openclaw\skills)
  ↓ 49 more

Selected: Amp, Cline, Codex +10 more
```

Captura 05. Se eligen los agentes que usarán las skills; en la selección aparece Codex.



Captura 06. Se selecciona Symlink como método de instalación.



Captura 07. Antes de continuar, se revisan las rutas, los agentes seleccionados y el resumen mostrado por el instalador.

```
C:\WINDOWS\system32\cmd.exe x + v

Details: https://skills.sh/tt-ali/archify
-----+-----
0 Proceed with installation?
  Yes
0 Installation complete

0 Installed 2 skills -----+-----
✓ ~\.agents\skills\archify
  universal: Amp, Cline, Codex, Cursor, Droid +7 more
  symlinked: CodeBuddy
✓ ~\.agents\skills\archify-review
  universal: Amp, Cline, Codex, Cursor, Droid +7 more
  symlinked: CodeBuddy
-----+-----

- Done! Review skills before use; they run with full agent permissions.

One-time prompt - you won't be asked again if you dismiss.

★ Install the find-skills skill? It helps your agent discover and suggest skills.
| > Yes / No
-
```

Captura 08. La consola confirma que archify y archify-review se instalaron en el directorio de skills del usuario.

```
C:\Windows\System32\cmd.exe x + v

0 Found 1 skill
• Selected 1 skill: find-skills

0 Installation Summary -----+-----
~\.agents\skills\find-skills
  universal: Amp, Cline, Codex, Cursor, Droid +7 more
  symlink → CodeBuddy
-----+-----

0 Security Risk Assessments -----+-----
find-skills  Gen      Socket      Snyk
             Safe      0 alerts    Med Risk
Details: https://skills.sh/vercel-labs/skills
-----+-----

0 Installation complete

0 Installed 1 skill -----+-----
✓ ~\.agents\skills\find-skills
  universal: Amp, Cline, Codex, Cursor, Droid +7 more
  symlinked: CodeBuddy
-----+-----
```

Captura 09. El asistente ofrece instalar find-skills y se confirma esta skill adicional.

# Petición a Codex para generar el mapa interactivo

## Prompt

### Versión en inglés para Archify

Use Archify to create a premium, modern, highly visual and interactive software architecture diagram for the following system.

### Core System

#### Mobile Client

Flutter Mobile App

Represent the mobile application as the primary user entry point.

Show requests from Flutter to authentication, backend APIs, databases and mapping services.

#### Authentication & Security Layer

Keycloak Authentication

Clearly visualize the authentication flow:

User

→ Flutter App

→ Keycloak

→ JWT / Access Token

→ Flutter App

→ FastAPI

Represent Keycloak as a dedicated security layer.

Show that protected FastAPI endpoints validate the JWT / access token before granting access to application resources.

#### Backend / API Layer

FastAPI REST API

Position FastAPI as the central communication and business-logic layer connecting:

Flutter

Keycloak

PostgreSQL

MongoDB

Maps services

Represent REST endpoints and, when possible, visually differentiate:

GET

POST

PUT / PATCH

DELETE

#### Data Layer

PostgreSQL

Represent structured and relational data such as:

Users

Roles

Devices

Entity relationships

Application configuration

MongoDB

Represent flexible, event-driven and geospatial information such as:

Geofences

Locations

Events

History

GeoJSON

Geospatial datasets

Clearly show which types of information FastAPI reads from and writes to each database.

## Maps & Geolocation

Leaflet / Maps Service

Show communication between Flutter, FastAPI and the mapping service.

Represent:

Maps

Markers

Locations

Coordinates

Geofences

Geographic visualization

Map tiles

GeoJSON

## Development Infrastructure

Docker

Represent application services as containers where appropriate:

FastAPI Container

PostgreSQL Container

MongoDB Container

Keycloak Container

Docker Compose

Represent Docker Compose as the local development orchestrator.

Show Docker Compose starting, networking and coordinating the different containers through an internal development network.

## Source Control

Git

Represent local source control.

GitHub

Represent:

Remote repository

Branches

Pull Requests

Collaboration

Potential CI/CD integration

Visualize the development flow:

Developer

→ Git

→ GitHub

→ Development / Deployment workflow

## Architecture Zones

Organize the architecture into clearly defined visual zones:

Client Layer

Authentication & Security Layer

API / Backend Layer

Data Layer

External Services

Development Infrastructure

Source Control

Use visual containers, grouped sections or boundaries to make each layer immediately recognizable.

## Trust Boundaries

Explicitly visualize trust boundaries between:

User device

Authentication services

Internal backend services

Databases

External mapping services

Development infrastructure

Use dashed boundaries, security zones or visually distinct containers.

## Main Data and Request Flows

Use clear, modern and, if possible, animated directional connections.

Authentication Flow

User

→ Flutter App

→ Keycloak

→ JWT / Access Token

→ Flutter App

→ FastAPI

API Request Flow

Flutter App

→ FastAPI

→ PostgreSQL / MongoDB

→ FastAPI

→ Flutter App

Geolocation Flow

Flutter App

→ FastAPI

→ MongoDB / Geospatial Data

→ Leaflet / Maps Service

→ Flutter App

Development Flow

Developer

→ Git

→ GitHub

→ Docker Compose

→ Application Services

## Connection Labels

Label important connections with their protocols or data formats.

Examples:

Flutter → Keycloak

OAuth 2.0 / OpenID Connect

Flutter → FastAPI

HTTPS / REST / JSON + Bearer Token

FastAPI → PostgreSQL

SQL

FastAPI → MongoDB

MongoDB Protocol / GeoJSON

Flutter / FastAPI → Maps Service

Map Tiles / GeoJSON / Coordinates

Git → GitHub

Push / Pull / Merge

## Visual Design

Create a premium futuristic technology architecture dashboard.

Use:

Dark-mode-first UI.

Deep dark background.

Glassmorphism cards.

Subtle gradients.

Soft shadows.

Neon glow accents.

Technology icons.

Distinct colors for each architecture layer.

Modern connection lines.

Subtle animated data-flow effects.

High visual hierarchy.

Clean typography.

Professional cloud architecture styling.

Visual inspiration:

GitHub

Vercel

Linear

Datadog

Grafana

AWS Architecture Diagrams

Modern DevOps / observability dashboards

Avoid making the result look like a static flowchart or basic organization chart.

## Interactivity

Generate the architecture as interactive HTML.

The diagram should support as many of the following interactions as Archify allows:

Zoom in / zoom out.

Canvas pan.

Drag nodes.

Hover states.

Technology tooltips.

Clickable architecture components.

Highlight connected dependencies when a node is selected.

Highlight upstream and downstream dependencies.

Toggle architecture layers.

Toggle request flows.

Show / hide infrastructure components.

Interactive legend.

Smooth transitions and animations.

## Node Detail Panel

When the user clicks a component, display a side panel or detail card containing:

Component name

Technology

Responsibility

Inputs

Outputs

Dependencies

Protocols

Data exchanged

Security considerations

Example:

FastAPI REST API

Responsibility:

Main backend and business logic.

Inputs:

HTTPS requests + JWT.

Outputs:

JSON responses.

Dependencies:

Keycloak, PostgreSQL, MongoDB and Maps Service.

Protocols:

HTTPS / REST.

## Interactive Flow Exploration

Allow the user to visually follow a complete request through the architecture.

For example, selecting:

"Authentication Flow"

should highlight only:

User

→ Flutter

→ Keycloak

→ JWT

→ Flutter

→ FastAPI

Selecting:

"Geolocation Flow"

should highlight:

Flutter

→ FastAPI

→ MongoDB

→ Maps Service

→ Flutter

Dim unrelated architecture components while a flow is selected.

## UX Requirements

The architecture should feel like an interactive observability and architecture-exploration tool rather than a static diagram.

The user should be able to understand at a glance:

What components exist.

How components communicate.

Which protocols are used.

Which data is exchanged.

Where authentication occurs.

Where trust boundaries exist.

Where persistent data is stored.

How an end-to-end request travels through the system.

How the local development environment is orchestrated.

## Final Output

Generate the architecture as a single interactive HTML experience.

The output should be:

Responsive.

Interactive.

Visually impressive.

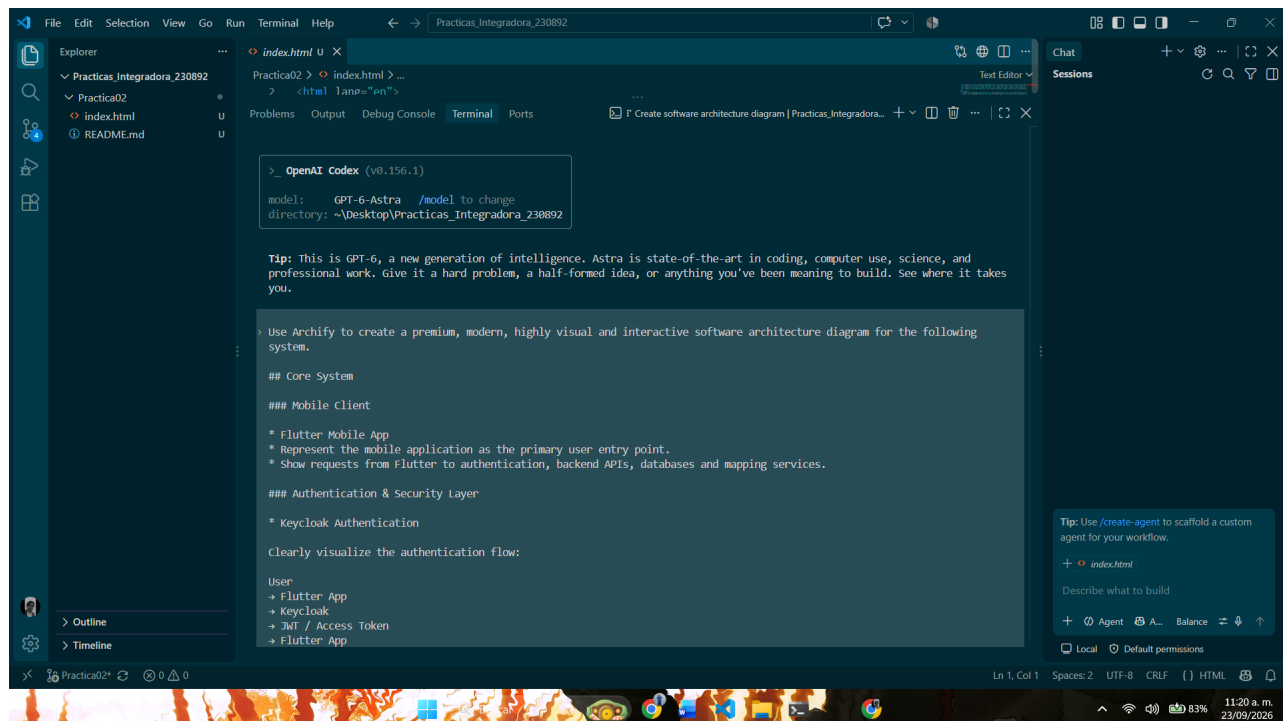
Easy to navigate.

Suitable for a software architecture presentation.

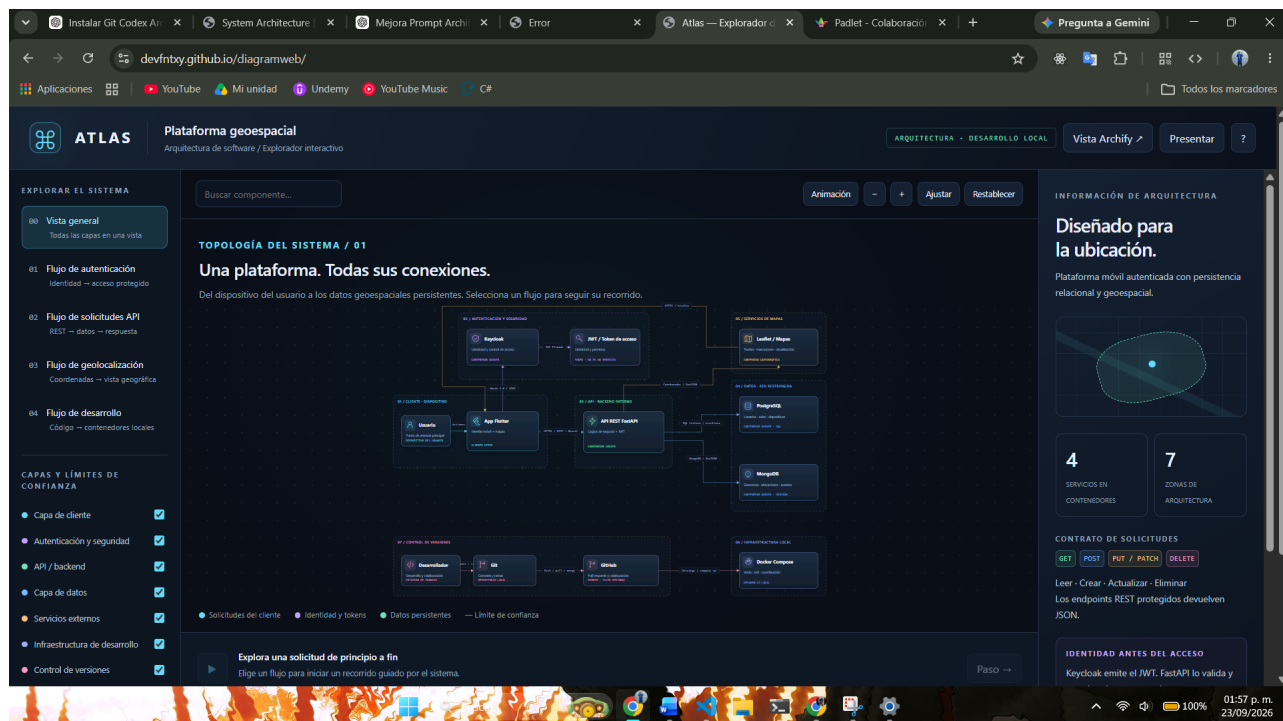
Suitable for technical documentation.

Suitable for displaying on a large screen during a project presentation.

Prioritize visual clarity, interaction and an impressive modern developer-tool aesthetic.



Captura 10. En VS Code se envía a Codex el prompt completo para crear el explorador interactivo de la arquitectura.



Captura 11. Se visualiza en el navegador el resultado ATLAS, con la topología del sistema y controles para explorar sus flujos y capas.